2.89 A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

(a) What is the probability that neither is available when needed?

(b) What is the probability that a fire engine is available when needed?

Solution:

P(A) = 0.96

P(B) = 0.96

Theorem 2.11: Two events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B).$$

Therefore, to obtain the probability that two independent events will both occur, we simply find the product of their individual probabilities.

(a) P (neither is available when needed?)

$P(A' \cap B')$=?

$\therefore P(A' \cap B') = P(A')\, P(B')$

$\therefore P(A') = 1 -$P(A)

$P(A' \cap B') = [1 - P(A)][1 - P(B)]$

$P(A' \cap B') = [1 - 0.96][1 - 0.96]$

$P(A' \cap B') = [0.04][0.04]$

$P(A' \cap B') = 0.0016$

(b) P (a fire engine is available when needed) ?

$P(A \cup B)$ =?

$\therefore P(A \cup B) = 1 - P(A' \cap B')$

$P(A \cup B) = 1 - 0.0016$

$P(A \cup B) = 0.9984$

2.92

Suppose the diagram of an electrical system is as given in Figure 2.10. What is the probability that the system works? Assume the components fail independently.

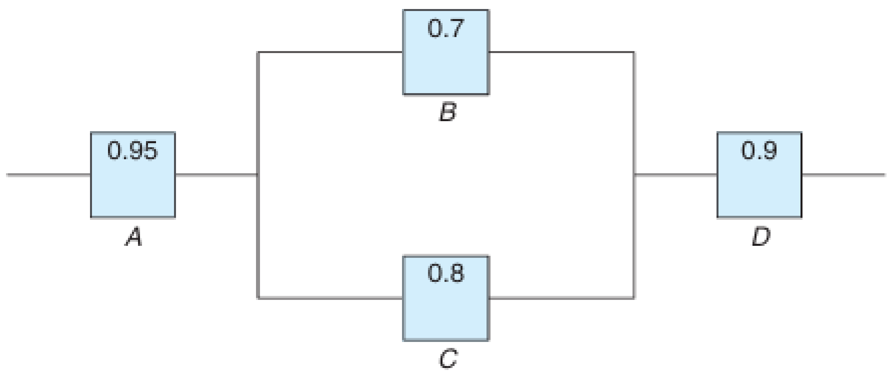
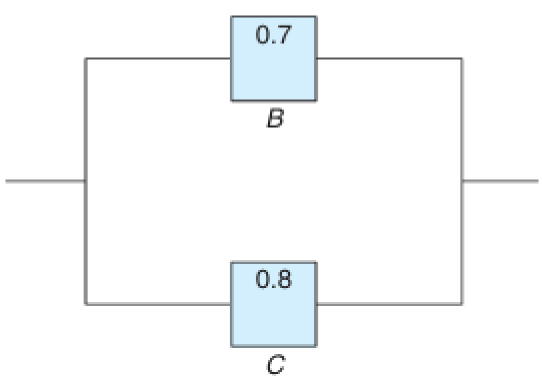


Figure 2.10: Diagram for Exercise 2.92.

Solution:

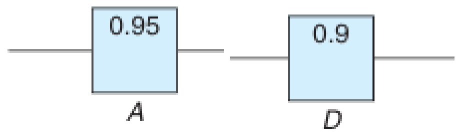
Tow types circuit

1-Parallal



$P(A \cup B)$

2-Series



$P(A \cap B)$

$P(A \cap (B \cup C) \cap D) = ?$

$P(A \cap (B \cup C) \cap D) = P(A)\ P(B \cup C)\ P(D)$

$\therefore P(A \cup B) = 1 - P(A' \cap B')$

$P(A \cap (B \cup C) \cap D) = P(A)\ [1 - P(B' \cap C')]\ P(D)$

$\therefore P(A' \cap B') = P(A')\ P(B')$

$P(A \cap (B \cup C) \cap D) = P(A)\ [1 - P(B')\ P(C')]\ \ P(D)$

$\therefore P(A \cup B) = 1 - P(A' \cap B')$

$P(A \cap (B \cup C) \cap D) = P(A)\ [1 - [1 - P(B)][1 - P(C)]\ ]\ \ P(D)$

$P(A \cap (B \cup C) \cap D) = (0.95)\ [1 - [1 - (0.7)][1 - (0.8)]\ ]\ \ (0.9)$

$P(A \cap (B \cup C) \cap D) = (0.95)\big[1 - [(0.3)][(0.2)]\big]\ (0.9)$

$P(A \cap (B \cup C) \cap D) = (0.95)(0.94)\ (0.9)$

$P(A \cap (B \cup C) \cap D) = \ 0.8037$

2.93

A circuit system is given in Figure 2.11. Assume the components fail independently.

(a) What is the probability that the entire system works?

(b) Given that the system works, what is the probability that the component A is not working?

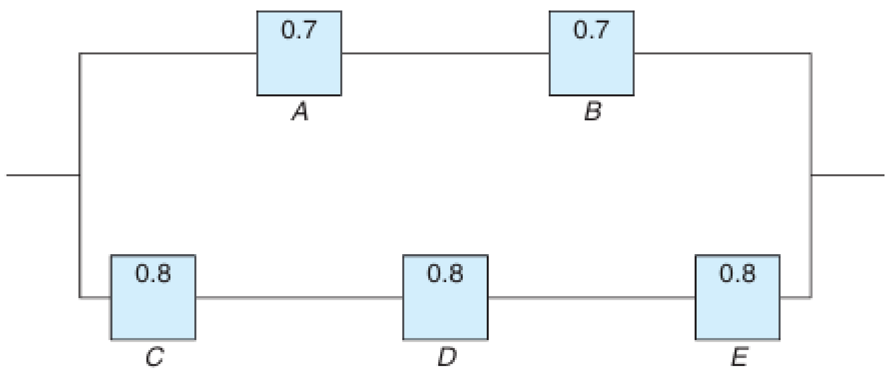


Figure 2.11: Diagram for Exercise 2.93.

Solution:

(a) P(the entire system works) ?

$P[(A \cap B) \cup (C \cap D \cap E)] = ?$

$\therefore P(A \cup B) = 1 - P(A' \cap B')$

$\therefore P(A') = 1 -$P(A)

$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - P[(A \cap B)' \cap (C \cap D \cap E)']$

$\therefore P(A' \cap B') = P(A') * P(B')$

$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - P[(A \cap B)'] * P[(C \cap D \cap E)']]$

$\therefore P(A') = 1 -$P(A)

$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - [1 - P\big[(A \cap B) \quad \big] * [1 - P[(C \cap D \cap E) \quad]]]$

$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - [1 - P(A) * P(B)] * [1 - P(C) * P(D) * P(E)]]$

$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - [1 - (0.7)(0.7)] * [1 - (0.8)(0.8)(0.8)]]$

$$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - [1 - 0.49] * [\, 1 - 0.512]]$$

$$P[(A \cap B) \cup (C \cap D \cap E)] = [1 - [0.51] * [\, 0.488]]$$

$$P[(A \cap B) \cup (C \cap D \cap E)] = \big[1 - [0.2488]\big] = 0.75112$$

(b) P(the component A is not working) ?

$$P = \frac{P(A' \cap C \cap D \cap E)}{P(System\ works)}$$

$$= \frac{(0.3)(0.8)(0.8)(0.8)}{0.75112} = 0.2045$$